

PERFORMANCE AND ECONOMIC ANALYSIS OF ELECTRIC VEHICLE CHARGING INTEGRATION IN THE WSSC 9-BUS SYSTEM

This project report is submitted to

Silicon Institute of Technology, Bhubaneswar

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

In

Electrical and Electronics Engineering

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MAY, 2026



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CERTIFICATE

This is to certify that the work contained in the project entitled **“PERFORMANCE AND ECONOMIC ANALYSIS OF ELECTRIC VEHICLE CHARGING INTEGRATION IN WSCC 9-BUS SYSTEM”**, submitted by **Hasina Fatema** (Regd no: 2201209286), **Laxmi Narayan Sahoo** (Regd no: 2201209306), **Priyanshu Sinha** (Regd no: 2201209336) and **Shaswat Kanungo** (Regd no: 2201209357) is a record of Bonafide works carried out by them under my supervision and guidance.

The contents embodied in the project is being submitted as a part of 8th semester project for the undergraduate curriculum and have not been submitted for the award of any other degree or diploma in this or any other university.

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DECLARATION

We hereby certify that: -

- a. The work contained in the project is original and has been done by ourselves under the supervision of our supervisor.
- b. The work has not been to any other Institute for any degree or diploma.
- c. We have confirmed to the norms and guidelines given to us by the Project Review Committee of our Department.
- d. Whenever we have used materials (data, theoretical analysis and text) from other sources, we have given due credit to them by citing them in the text of the project and giving their details in the references.

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ACKNOWLEDGEMENT

We are highly indebted to Prof. Rajan Kumar Mishra, our supervisor and mentor, for his valuable guidance, constant support throughout the completion of project. His suggestions and expert knowledge helped us greatly in carrying out this work successfully.

We would like to express our sincere thanks to our college and the Department of Electrical and Electronics Engineering for providing us with this opportunity to enhance our knowledge and technical skills through this project.

We would also like to convey our heartfelt gratitude to all our respected teachers who guided and inspired us at every stage of this journey.

Finally, we would like to thank our parents, classmates, and Almighty God for their blessings, motivation, and support in helping us achieve our goals.

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ABSTRACT

The growing adoption of Electric Vehicles (EVs) adds significant load demand to modern power systems, making the integration of EV charging infrastructure an important planning consideration. Improper placement or sizing of charging stations can increase power losses, voltage deviations, and operational costs, thereby affecting the overall efficiency and economics of the system. This project presents a combined technical and economic analysis of EV charging integration in the WSCC 9-bus test system using Power Flow (PF) analysis.

A base case load flow study is first carried out on the standard WSCC 9-bus system to establish reference operating conditions. The EV charging station is modelled as an additional load with specified active power demand, while the optimized reactive power requirement (Q_{EV}) is determined using the **Particle Swarm Optimization (PSO)** technique. In this study, the maximum EV charging penetration level is considered as **30% (0.3 p.u.)** of the connected load. Three different cases are analysed by placing the EV charging station at different load buses while maintaining identical transmission line parameters. For each case, power flow analysis is performed to evaluate voltage profile, transmission losses, and generation cost.

The results demonstrate that the placement of the EV charging station has a significant influence on both technical performance and economic operation of the system. The study identifies the most suitable bus location based on improved voltage stability, reduced power losses, and minimized total generation cost. The use of PSO for optimized Q_{EV} calculation further enhances system performance and provides valuable insights for effective and cost-efficient EV charging infrastructure planning in modern power systems.

Keywords: *Electric Vehicle (EV), WSCC 9-Bus System, Power Flow Analysis, Newton-Raphson Method, Particle Swarm Optimization.*

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1.Introduction

The rapid growth of Electric Vehicles (EVs) has become a major step toward sustainable transportation and reduction of environmental pollution caused by conventional fuel-based vehicles. Governments and industries across the world are promoting EV adoption through supportive policies, charging infrastructure development, and technological advancements. However, the increasing number of EVs creates additional load demand on existing electrical power systems, especially during peak charging periods. This makes the integration of EV charging stations an important challenge for power system planners and utilities.

EV charging stations, when connected without proper planning, can adversely affect the performance of the power network. Additional charging demand may cause voltage deviations at load buses, increase transmission line losses, overload network components, and raise overall generation cost. Therefore, identifying the most suitable location and proper sizing of EV charging infrastructure is essential to maintain reliable and economical system operation.

Power Flow (PF) analysis is one of the most effective methods for studying the steady-state behavior of power systems under different loading conditions. It provides important information such as bus voltages, real and reactive power flows, and transmission losses. In this project, the standard WSCC 9-Bus test system is selected as the study network because it is widely used for research and educational purposes. A base case load

flow study is first carried out to determine the normal operating condition of the system before EV charging integration.

The EV charging station is modeled as an additional load with specified active power demand. Since reactive power support also plays a significant role in maintaining voltage stability and reducing losses, the reactive power requirement of EV charging (QEV) is optimized using the Particle Swarm Optimization (PSO) technique. PSO is a population-based optimization method inspired by the social behavior of bird flocking and is widely used for solving engineering optimization problems.

In this study, the maximum EV charging penetration level is considered as 30% (0.3 p.u.) of the connected load. Different charging placement cases are analyzed by connecting the EV load at selected buses while keeping transmission system parameters constant. For each case, Newton-Raphson load flow analysis is performed to evaluate voltage profile, power losses, and economic operation.

The main objective of this work is to determine the most suitable bus location for EV charging integration that provides improved voltage stability, minimum transmission losses, and reduced generation cost. The results of this study can be useful for future EV infrastructure planning and smart grid development with efficient charging management strategies.

2.Literature Review

AUTHOR	PUBLISHER/SOURCE	YEAR	ABSTRACT
F.Schafer and M.Braun	IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)	2018	This paper presents an efficient open-source implementation for computing the Jacobian matrix used in the Newton-Raphson power flow algorithm. The study improves computational efficiency and speed of load flow calculations, making Newton-Raphson more suitable for large-scale modern power systems.
M. R. Hossain, M. S. Kaiser, F. I. Ali and M. M. A. Rizvi	International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)	2015	The authors proposed network flow optimization using Genetic Algorithm combined with Newton-Raphson load flow analysis. The work demonstrates that

			optimization techniques can reduce system losses and improve power system operation under varying load conditions.
W. Chuanjun, W. Ling and R. Xuejing	IEEE 2nd International Conference on Electrical Engineering, Big Data and Algorithms (EEBDA)	2023	This paper discusses the general Particle Swarm Optimization (PSO) algorithm and its engineering applications. PSO is shown to be effective for solving nonlinear optimization problems with faster convergence and better global search capability.
Hadi Saadat	McGraw-Hill, <i>Power System Analysis</i> (3rd Edition)	2010	This textbook provides detailed concepts of power flow analysis, Y-bus formation, Newton-Raphson method, economic dispatch, and system stability. It is widely used as a standard reference for power system engineering studies.
J.J. Grainger and W.D. Stevenson	McGraw-Hill, <i>Power System Analysis</i>	1994	This book explains fundamental principles of

			power system modeling, load flow studies, transmission system analysis, and operational planning. It serves as a valuable source for understanding practical power system problems and solution methods.
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3. Electric Vehicle and Charging Integration

The rapid growth of Electric Vehicles (EVs) has created a major transformation in both the transportation and electrical power sectors. EVs are environmentally friendly alternatives to conventional fuel-based vehicles because they reduce carbon emissions, lower air pollution, and improve energy efficiency. However, the increasing number of EVs also creates additional demand on the electrical grid because these vehicles require regular charging. Therefore, EV charging integration has a significant impact on power system planning, operation, and stability.

When EV charging stations are connected to the grid, they behave as new electrical loads. Depending on the number of vehicles charging simultaneously, charging power level, and location of charging stations, the effect on the power system may be either beneficial or challenging. If charging demand is not managed properly, it can create technical and economic problems. Hence, understanding the impact of EV charging integration is essential.

1. Increase in Load Demand

The most immediate impact of EV charging is the increase in total system load. Every EV charger draws power from the grid to charge the battery. When many vehicles charge at the same time, especially during evening hours, the total demand rises significantly.

The total load after EV integration can be represented as:

$$P_{Total} = P_{Base} + P_{EV}$$

Where:

- P_{Base} = Existing system demand
- P_{EV} = EV charging demand

This additional load can stress feeders, transformers, and transmission lines.

2. Peak Load Increase

Many EV users prefer charging after returning home in the evening. This charging time often coincides with normal residential peak demand. As a result, the peak load becomes higher than usual.

Effects of peak demand increase:

- Overloading of transformers
- Increased generation requirement
- Higher operating cost
- Reduced reserve margin
- Need for infrastructure upgrade

If unmanaged, this can reduce system reliability.

3. Voltage Drop in Network

When large charging stations are connected at weak buses or remote locations, high current flows through transmission or distribution lines. This causes voltage drop across line impedance.

$$V_{drop} = IZ$$

Consequences of voltage drop:

- Poor voltage profile
- Low voltage at consumer end
- Reduced equipment performance
- Need for reactive power compensation

Thus, charging station location is very important.

4. Increase in Power Losses

As line current increases due to EV charging, copper losses in conductors also increase.

$$P_{loss} = I^2 R$$

This means:

- More charging demand causes more current
- Higher current increases line losses
- Overall system efficiency decreases

Hence, optimization of EV charging location and timing is required to minimize losses.

5. Reactive Power Impact

Power electronic converters used in EV chargers may absorb or generate reactive power depending on charger design. Improper reactive power balance can affect voltage regulation and system stability.

Effects include:

- Voltage fluctuation
- Lower power factor
- Increased current flow
- Additional transmission losses

In this project, EV charging was represented as reactive power load and optimized using PSO.

6. Transformer and Equipment Overloading

Distribution transformers are generally designed for conventional loads. Sudden large-scale EV charging may exceed their rating.

Possible problems:

- Transformer overheating

- Reduced insulation life
- Frequent maintenance
- Unexpected failures

Therefore, utilities must reinforce infrastructure before large EV penetration.

In the present project, EV charging loads were integrated into the WSCC 9-bus system at buses 5, 6, and 8. It was observed that charging location directly affects system losses. Therefore, Particle Swarm Optimization was used to allocate charging reactive power optimally.

The optimized charging distribution improved system performance by selecting buses that gave lower losses and better operating conditions.

Electric Vehicle charging integration has both positive and negative impacts on power systems. Uncontrolled charging may increase load demand, power losses, voltage drop, and equipment stress. However, with proper planning, optimization, and smart charging control, EV integration can improve system efficiency and support future smart grid development. This project confirms that optimized EV charging allocation is an effective method for reducing adverse impacts and ensuring reliable power system operation.

4. WSCC 9-Bus Test System

The WSCC 9-Bus Test System is a standard benchmark model widely used in power system studies for analyzing load flow, stability, fault analysis, and optimization techniques. WSCC stands for **Western Systems Coordinating Council**, and this system represents a simplified practical transmission network containing generators, loads, transformers, and transmission lines. Because of its moderate size and clear structure, it is commonly used in academic research and simulation software such as MATLAB, ETAP, PSCAD, and PowerWorld.

The system consists of **9 buses**, **3 generator buses**, **3 load buses**, and **9 transmission lines/transformer branches**. It is designed to represent the operation of a small interconnected power grid. The network includes one slack bus, two PV buses, and six PQ buses.

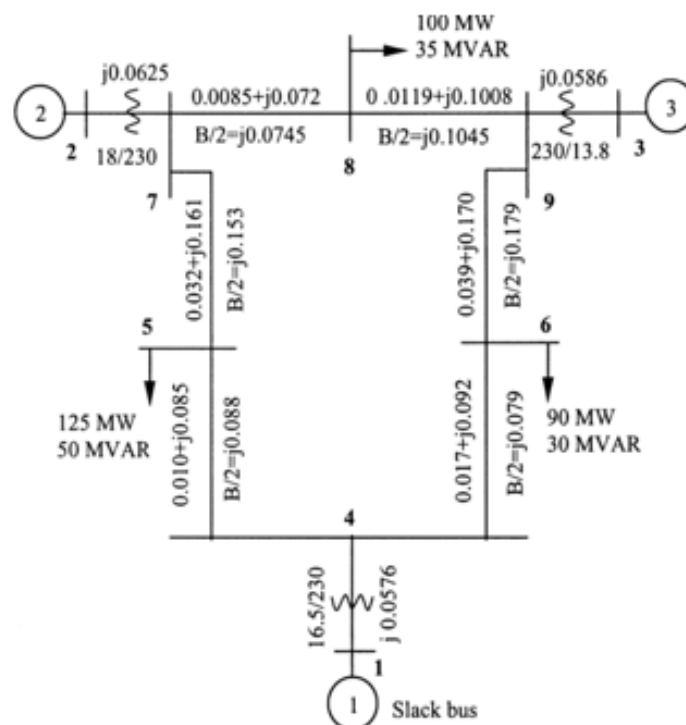


Fig 1: WSCC 9-Bus Test System Single Line Diagram

Bus Classification

The buses in the WSCC 9-bus system are categorized as:

- **Bus 1** – Slack Bus
- **Bus 2 and Bus 3** – PV (Generator) Buses
- **Bus 4 to Bus 9** – PQ (Load/Network) Buses

The slack bus maintains the system voltage magnitude and balances active and reactive power mismatch in the network. PV buses control generator real power output and voltage magnitude, while PQ buses represent load buses where real and reactive power demands are specified.

Generation and Load Data

The generators are connected at buses 1, 2, and 3. These generating units supply real power to the system and regulate voltage. Major loads are connected at buses 5, 6, and 8, where power demand is specified in MW and MVAR.

From the given data:

- Bus 5 load = 125 MW and 50 MVAR
- Bus 6 load = 90 MW and 30 MVAR
- Bus 8 load = 100 MW and 35 MVAR

These loads create power demand that must be supplied through the transmission network.

Line Data Overview

The **line data matrix** defines the electrical parameters of each branch connecting buses. It contains:

- Sending bus number
- Receiving bus number

- Resistance (R)
- Reactance (X)
- Line charging susceptance (B/2)
- Tap ratio

Example:

4 5 0.01 0.085 0.088 1

This means bus 4 is connected to bus 5 with specified resistance, reactance, charging susceptance, and unity tap ratio.

Transmission lines in the system transfer electrical power from generation buses to load buses. Their impedance affects voltage drops, line currents, and power losses.

Bus Data Overview

The bus data matrix stores the operating parameters of each bus such as:

- Bus number
- Generated real/reactive power
- Load real/reactive power
- Voltage magnitude
- Voltage angle
- Bus type

Example:

1 0 0 0 0 1.04 0 1

This indicates Bus 1 is the slack bus with voltage magnitude of 1.04 per unit.

The bus data is essential for load flow studies because it defines system operating conditions and known/unknown variables.

Y-Bus Matrix Formation

The provided MATLAB code forms the **Bus Admittance Matrix (Y-Bus)** using line data. Y-Bus is the most important matrix in power flow analysis because it mathematically represents the complete network.

Each diagonal element gives self-admittance of a bus, while off-diagonal elements give mutual admittance between connected buses.

The general matrix form is:

$$I = Y_{bus}V$$

Where:

- **I** = Bus current vector
- **Y_{bus}** = Bus admittance matrix
- **V** = Bus voltage vector

The MATLAB function reads each line branch, calculates branch admittance, and updates matrix elements accordingly.

Purpose of Y-Bus:

- Used in load flow calculations
- Used in fault studies
- Used in stability analysis
- Determines bus currents and voltages

```

%% =====
% YBUS
%% =====
function [yb,nbus] = ybus1(linedata)

j = sqrt(-1);

nbus=max(max(linedata(:,1)),max(linedata(:,2)));
nbr=size(linedata,1);

z=linedata(:,3)+j*linedata(:,4);
y=1./z;

yb=zeros(nbus,nbus);

for i=1:nbr

    from=linedata(i,1);
    to=linedata(i,2);
    tap=linedata(i,6);
    b=linedata(i,5);

    yb(from,from)=yb(from,from)+y(i)/(tap^2)+j*b;
    yb(to,to)=yb(to,to)+y(i)/(tap^2)+j*b;

    yb(from,to)=yb(from,to)-y(i)/tap;
    yb(to,from)=yb(from,to);

end

end

```

Fig 2: Matlab code of Y_{bus} formation

Importance of WSCC 9-Bus System

The WSCC 9-bus system is widely used because:

- It is simple and easy to understand
- Contains all practical bus types
- Suitable for testing load flow algorithms
- Useful for Newton-Raphson and Gauss-Seidel methods
- Helps study renewable and EV integration
- Standard benchmark for research comparison

It provides realistic behavior while remaining computationally manageable.

```

%% ===== LINE DATA =====
linedata = [
    1 4 0      0.0576 0      1
    2 7 0      0.0625 0      1
    3 9 0      0.0586 0      1
    4 5 0.01    0.085   0.088 1
    4 6 0.017   0.092   0.079 1
    5 7 0.032   0.161   0.153 1
    6 9 0.039   0.17    0.179 1
    7 8 0.0085  0.072   0.0745 1
    8 9 0.0119  0.1008  0.1045 1
];

```

```

%% ===== BUS DATA =====
busdata = [
    1 0      0 0      0      1.04 0 1 1      -1
    2 1.63 0 0      0      1.025 0 3 0.06 -1
    3 0.85 0 0      0      1.025 0 3 0.5  -0.1
    4 0      0 0      0      1      0 2 0      0
    5 0      0 1.25 0.5  1      0 2 0      0
    6 0      0 0.9  0.3  1      0 2 0      0
    7 0      0 0      0      1      0 2 0      0
    8 0      0 1      0.35 1      0 2 0      0
    9 0      0 0      0      1      0 2 0      0
];

```

Fig 3: Matlab Code for bus data and line data

Application with Newton-Raphson Load Flow

In this project, the WSCC 9-bus system is used as the test network for Newton-Raphson load flow analysis. The bus data and line data are provided as inputs, then the Y-Bus matrix is formed. After that, voltage magnitude, voltage angle, power flow, and system losses are calculated iteratively.

This makes the system ideal for evaluating effects of additional EV charging loads, reactive power compensation, and optimization methods.

The WSCC 9-Bus Test System is a standard small-scale power network used for simulation and performance analysis. It includes generators, loads, transformers, and transmission lines arranged in a realistic

interconnected structure. Using the provided bus data, line data, and Y-Bus formation code, accurate load flow analysis can be performed. It is highly suitable for educational, research, and smart grid studies involving Newton-Raphson techniques.

5. Load Flow Analysis

Load flow analysis is one of the most important studies in power system engineering. It is used to determine the steady-state operating condition of a power system network under normal operating conditions. The main objective of load flow analysis is to calculate the voltage magnitude, phase angle, real power, reactive power, and power losses at different buses in the system. It helps engineers understand how electrical power flows from generating stations to load centers through transmission lines.

In a power system, buses are classified into three types: slack bus, PV bus, and PQ bus. The slack bus is used as a reference bus where voltage magnitude and angle are specified. The PV bus represents generator buses where real power and voltage magnitude are known, while reactive power and voltage angle are calculated. The PQ bus represents load buses where real and reactive powers are specified, and voltage magnitude and angle are unknown.

Load flow studies are widely used for planning, operation, and control of power systems. It helps in determining overloaded lines, maintaining voltage limits, minimizing transmission losses, and improving system reliability. It is also useful for expansion planning and integrating renewable energy sources such as solar photovoltaic systems and wind energy systems into the grid.

Several numerical methods are used to solve load flow problems such as Gauss-Seidel method, Newton-Raphson method, and Fast Decoupled method. Among these methods, the Newton-Raphson method is the most widely used because of its high accuracy and fast convergence for large systems.

5.1 Newton-Raphson Method for Load Flow

The Newton-Raphson (NR) method is an iterative numerical technique used to solve nonlinear algebraic equations in power flow studies.

Since load flow equations are nonlinear in nature, the NR method is highly suitable for solving them efficiently.

The method starts with an initial guess of voltage magnitude and voltage angle at all buses. Then power mismatches between specified values and calculated values are determined. These mismatches are reduced iteratively using the Jacobian matrix until the error becomes very small.

The basic equation of Newton-Raphson load flow is:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J \begin{bmatrix} \Delta \delta \\ \Delta V \end{bmatrix}$$

Where:

- ΔP = Real power mismatch
- ΔQ = Reactive power mismatch
- J = Jacobian matrix
- $\Delta \delta$ = Change in bus angle
- ΔV = Change in bus voltage magnitude

The Jacobian matrix consists of four submatrices:

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

The iteration process continues until the mismatch values become less than the specified tolerance.

Advantages of Newton-Raphson Method:

- Fast convergence speed
- High accuracy
- Suitable for large power systems
- Less number of iterations compared to Gauss-Seidel method

- Reliable under varying loading conditions

```

%% =====
% NR LOAD FLOW FUNCTION
%% =====
function Total_Loss = NR_loadflow(busdata, linedata, Qval)

j = sqrt(-1);

% Add EV reactive loads simultaneously
busdata(5,5) = busdata(5,5) + Qval(1);
busdata(6,5) = busdata(6,5) + Qval(2);
busdata(8,5) = busdata(8,5) + Qval(3);

[ybus,nbus] = ybus1(linedata);

vm = busdata(:,6);
delta = busdata(:,7);

dpq=zeros(1,nbus);
dpv=dpq;
dns=dpq;

dpq(busdata(:,8)==2)=1;
dpv(busdata(:,8)==3)=1;
dns(busdata(:,8)>1)=1;

psh = busdata(:,2)-busdata(:,4);
qsh = busdata(:,3)-busdata(:,5);

iter=1;
maxerror=100;

while maxerror>1e-4 && iter<200

    v = vm.*cos(delta)+j*vm.*sin(delta);
    s = v.*conj(ybus*v);

    pcalc = real(s);
    qcalc = imag(s);

    delp = psh-pcalc;
    delq = qsh-qcalc;

    dc = [delp(dns==1);delq(dpq==1)];

```

Fig 4.1: Matlab Code for Newton Raphson

```

Jx = ybus.*repmat(v.',nbus,1).*repmat(conj(v),1,nbus);

J11 = -imag(Jx);
J11 = J11-diag(sum(J11,2));
J11 = J11(dns==1,dns==1);

J12 = real(Jx);
J12 = (J12+diag(sum(J12,2)))./repmat(vm',nbus,1);
J12 = J12(dns==1,dpq==1);

J21 = -real(Jx);
J21 = J21-diag(sum(J21,2));
J21 = J21(dpq==1,dns==1);

J22 = -imag(Jx);
J22 = (J22+diag(sum(J22,2)))./repmat(vm',nbus,1);
J22 = J22(dpq==1,dpq==1);

J = [J11 J12;J21 J22];

delv = J\dc;

edit = [dns';dpq'];
edit(edit==1)=delv;

delta = delta + edit(1:nbus);
vm = vm + edit(nbus+1:end);

iter = iter + 1;
maxerror = max(abs(delv));

end

%% LOSS
y = 1./(linedata(:,3)+1j*linedata(:,4));
Vn = vm.*(cos(delta)+1j*sin(delta));

Iline = (Vn(linedata(:,1))-Vn(linedata(:,2))).*y;
Sl = (abs(Iline).^2)./y;

Total_Loss = sum(real(Sl))*100;

end

```

Fig 4.2: Matlab Code for Newton Raphson

The provided MATLAB code is developed to perform load flow analysis using the Newton-Raphson (NR) method and determine the total real power loss in the power system network. It is designed to analyze the system under base case conditions as well as under additional electric vehicle (EV) reactive loading conditions.

At the beginning, the program calculates the **base case loss** by calling the NR load flow function with zero additional reactive load values. This represents the normal operating condition of the system before EV charging loads are connected.

```
BaseLoss = NR_loadflow(busdata, linedata, [0 0 0]);
```

The main function then takes bus data, line data, and reactive power values as inputs. The reactive loads are added to selected buses, representing simultaneous EV charging demand connected at multiple locations in the system.

Next, the code forms the **bus admittance matrix (Y-bus)** from the transmission line data. This matrix is essential in load flow analysis because it represents the electrical connections and impedances between buses.

The Newton-Raphson iterative process then begins using initial voltage magnitudes and phase angles. In each iteration, the program calculates the real and reactive power at each bus, compares them with the specified values, and determines the mismatch errors.

To minimize these errors, the program constructs the **Jacobian matrix**, which relates power mismatches to voltage magnitude and angle corrections.

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J \begin{bmatrix} \Delta \delta \\ \Delta V \end{bmatrix}$$

Using this relation, updated voltage values are obtained repeatedly until the mismatch error falls below the selected tolerance level. This indicates convergence of the solution.

After convergence, the code calculates current flowing through each transmission line and determines the corresponding real power losses. The sum of all line losses gives the **total system power loss**.

Finally, the function returns the total loss value, which can be compared with the base case loss to study the impact of EV charging loads on system performance.

Overall, the code provides an efficient method to:

- Perform Newton-Raphson load flow analysis
- Study steady-state bus voltages and angles
- Include additional EV reactive loads
- Calculate total transmission power loss
- Compare normal and loaded operating conditions

This approach is highly useful in modern smart grid and EV integrated power system studies.

6. Mathematical Modelling and Equations

The proposed work combines Newton-Raphson load flow analysis with Particle Swarm Optimization (PSO) for minimizing transmission power loss in the WSCC 9-bus system under EV reactive loading conditions. Several mathematical equations are used for modeling power flow, voltage calculations, loss estimation, and optimization. These equations form the foundation of the complete system.

1. Bus Current Equation

In power systems, the relationship between bus current and bus voltage is represented using the bus admittance matrix:

$$I = Y_{bus}V$$

Where:

- I = Bus current vector
- Y_{bus} = Bus admittance matrix
- V = Bus voltage vector

This equation is used to represent the electrical behavior of the network.

2. Complex Power Equation

The complex power injected at each bus is:

$$S_i = P_i + jQ_i = V_i I_i^*$$

Where:

- P_i = Real power
- Q_i = Reactive power
- V_i = Bus voltage
- I_i^* = Conjugate of bus current

This equation is used to calculate active and reactive power at each bus.

3. Real Power Flow Equation

The real power injected at bus i is given by:

$$P_i = \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \cos(\theta_{ij} + \delta_j - \delta_i)$$

Where:

- $|V_i|, |V_j|$ = Voltage magnitudes
- Y_{ij} = Admittance between buses
- δ_i, δ_j = Voltage angles

This equation is nonlinear and solved iteratively using Newton-Raphson method.

4. Reactive Power Flow Equation

Reactive power at bus i is:

$$Q_i = - \sum_{j=1}^n |V_i| |V_j| |Y_{ij}| \sin(\theta_{ij} + \delta_j - \delta_i)$$

Reactive power is important for maintaining voltage stability in the network.

5. Power Mismatch Equations

During load flow analysis, mismatches are calculated as:

$$\Delta P_i = P_i^{spec} - P_i^{calc}$$

$$\Delta Q_i = Q_i^{spec} - Q_i^{calc}$$

Where:

- Specified power = Given system data
- Calculated power = Obtained from present iteration

The objective of Newton-Raphson method is to reduce these mismatches to zero.

6. Newton-Raphson Matrix Equation

The correction equation used in each iteration is:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J \begin{bmatrix} \Delta \delta \\ \Delta V \end{bmatrix}$$

Where:

- J = Jacobian matrix
- $\Delta \delta$ = Change in voltage angle
- ΔV = Change in voltage magnitude

Updated values are:

$$\delta^{new} = \delta^{old} + \Delta \delta$$

$$V^{new} = V^{old} + \Delta V$$

This process continues until convergence.

7. Transmission Line Current Equation

Current flowing through a line connecting buses i and j is:

$$I_{ij} = (V_i - V_j)Y_{ij}$$

This equation is used to determine branch currents after load flow convergence.

8. Power Loss Equation

The real power loss in each transmission line is:

$$P_{loss} = I_{ij}^2 R_{ij}$$

Total system loss is:

$$P_{Total} = \sum P_{loss}$$

This is the main objective function minimized using PSO.

9. Reactive Power Constraint Equation

In the proposed optimization problem:

- Q5 and Q6 are decision variables
- Q8 is determined from total available reactive load

$$Q_8 = 0.3 - Q_5 - Q_6$$

Constraint:

$$Q_5 + Q_6 + Q_8 = 0.3$$

This ensures total EV reactive demand remains constant.

10. PSO Velocity Update Equation

Particle velocity is updated as:

$$v_i^{k+1} = wv_i^k + c_1r_1(pbest_i - x_i^k) + c_2r_2(gbest - x_i^k)$$

Where:

- w = inertia weight
- c1 = cognitive coefficient
- c2 = social coefficient
- r1, r2 = random numbers

11. PSO Position Update Equation

New particle position is:

$$x_i^{k+1} = x_i^k + v_i^{k+1}$$

Particles move toward the optimum reactive power allocation.

12. Objective Function

The optimization target is:

$$\text{Minimize } f(x) = P_{Total Loss}$$

Thus, PSO searches for the values of Q_5 , Q_6 , and Q_8 that give minimum transmission loss.

The proposed system uses mathematical equations from power flow theory and optimization techniques. Newton-Raphson equations calculate bus voltages, currents, and power flow, while PSO equations search for the best reactive power allocation. Together, these equations help reduce transmission losses, improve voltage regulation, and enhance the performance of the WSCC 9-bus power system.

7. Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) is a population-based optimization technique inspired by the social behavior of birds flocking and fish schooling. It is widely used in engineering applications because of its simple implementation, fast convergence, and ability to solve nonlinear optimization problems. In this project, PSO is used to determine the optimal reactive power distribution for electric vehicle (EV) charging loads connected to the WSCC 9-bus power system.

The main objective of applying PSO in this work is to minimize total transmission power loss by selecting the best reactive power values at selected buses. Since the relationship between reactive power loading and system loss is nonlinear, PSO becomes an effective optimization tool.

Objective of PSO in This System

The optimization variables considered are:

- **Q5** = Reactive power at Bus 5
- **Q6** = Reactive power at Bus 6
- **Q8** = Remaining reactive power at Bus 8

The constraint used in the system is:

$$Q_8 = 0.3 - Q_5 - Q_6$$

This means the total reactive load is fixed at 0.3 p.u., and PSO distributes it optimally among buses 5, 6, and 8.

The optimized values are passed to the Newton-Raphson load flow program, and the resulting total power loss is calculated. The PSO algorithm repeatedly searches for the combination that gives minimum loss.

Working Principle of PSO

In PSO, each possible solution is called a **particle**. A group of particles forms the swarm. Each particle has:

- Position (candidate solution)
- Velocity (movement direction)
- Personal best position (**pbest**)
- Global best position (**gbest**)

The particles move through the search space and continuously improve their positions using experience from themselves and the best particle in the swarm.

The velocity update equation is:

$$v_i^{k+1} = wv_i^k + c_1r_1(pbest_i - x_i^k) + c_2r_2(gbest - x_i^k)$$

The position update equation is:

$$x_i^{k+1} = x_i^k + v_i^{k+1}$$

Where:

- **w** = inertia weight
- **c1** = cognitive coefficient
- **c2** = social coefficient
- **r1, r2** = random numbers

```

%% =====
% PSO SETTINGS
%% =====

nPop = 50;
MaxIter = 40;

% Decision variables:
% x(1)=Q5 , x(2)=Q6
% Q8 = 0.3 - Q5 - Q6

lb = 0;
ub = 0.3;

%% ===== INITIAL POPULATION =====
x = zeros(2,nPop);

for i = 1:nPop
    q5 = rand*0.3;
    q6 = rand*(0.3-q5);
    x(:,i) = [q5;q6];
end

v = zeros(size(x));

pbest = x;
gbest = x(:,1);

BestLoss = zeros(MaxIter,1);

%% ===== INITIAL GLOBAL BEST =====
vals = zeros(1,nPop);
for i = 1:nPop
    vals(i) = fx(x(:,i),busdata,linedata);
end
[~,idx] = min(vals);
gbest = x(:,idx);

```

Fig 5.1: Matlab Code for PSO

```

%% =====
% PSO MAIN LOOP
%% =====

for it = 1:MaxIter

    w = 0.9 - (0.5*it/MaxIter);
    c1 = 2;
    c2 = 2.5;

    for i = 1:nPop

        v(:,i) = w*v(:,i) ...
            + c1*rand*(pbest(:,i)-x(:,i)) ...
            + c2*rand*(gbest-x(:,i));

        x(:,i) = x(:,i) + v(:,i);

        % limits
        x(x<lb)=lb;
        x(x>ub)=ub;

        % enforce Q5+Q6 <= 0.3
        if sum(x(:,i)) > 0.3
            x(:,i)=0.3*x(:,i)/sum(x(:,i));
        end

        % update pbest
        if fx(x(:,i),busdata,linedata) < fx(pbest(:,i),busdata,linedata)
            pbest(:,i)=x(:,i);
        end
    end

    % update gbest
    vals = zeros(1,nPop);
    for i = 1:nPop
        vals(i)=fx(x(:,i),busdata,linedata);
    end

    [BestLoss(it),idx]=min(vals);
    gbest=x(:,idx);

end

```

Fig 5.2: Matlab Code for PSO

8. MATLAB Methodology

The proposed work is implemented in **MATLAB** to analyze the WSCC 9-bus power system using **Newton-Raphson load flow** and **Particle Swarm Optimization (PSO)**. MATLAB is selected because of its strong numerical computation capability, matrix handling features, and easy simulation environment. The methodology includes data preparation, Y-bus formation, base case load flow analysis, optimization process, and result evaluation.

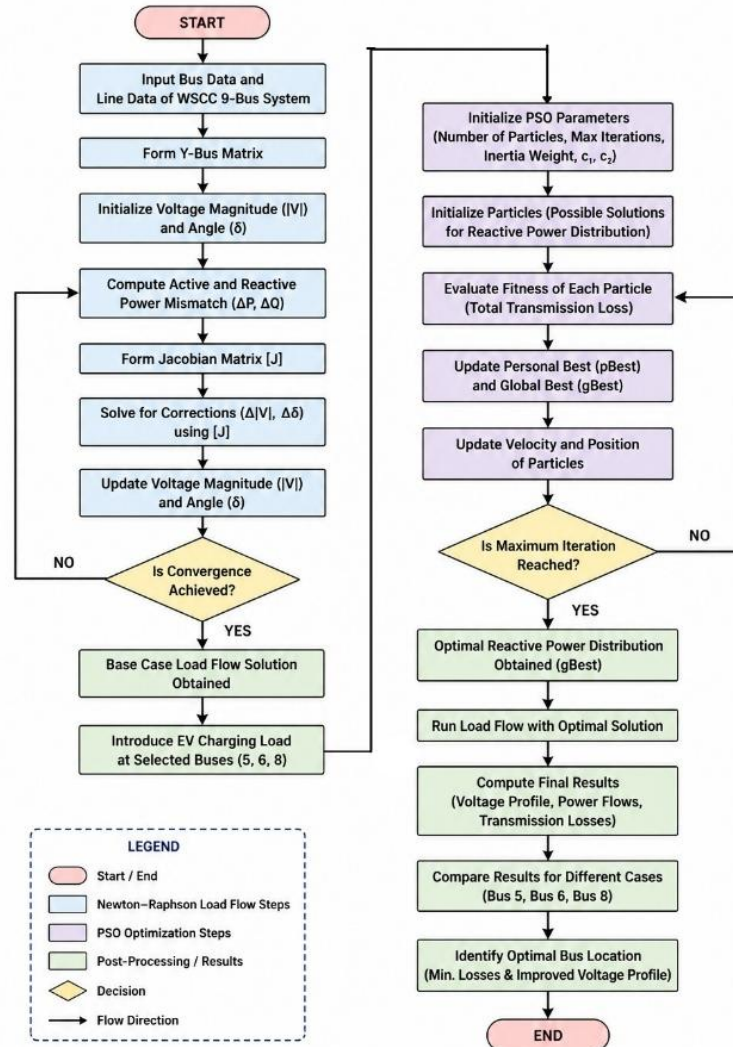


Fig 6: Newton Raphson Load Flow with PSO optimization

Stepwise Methodology

- Define WSCC 9-bus system bus data and line data
- Develop Y-bus admittance matrix using network parameters
- Initialize voltage magnitude and angle values
- Perform Newton-Raphson load flow analysis
- Calculate base case transmission power loss
- Define optimization variables for EV reactive loads
- Initialize PSO parameters and particle population
- Evaluate fitness using NR load flow function
- Update particle velocity and position iteratively
- Apply system operating constraints
- Obtain optimal reactive power distribution
- Calculate minimized system power loss
- Compare optimized result with base case
- Analyze convergence and system performance

i. Input Data Preparation

The first step in MATLAB implementation is entering the **bus data** and **line data** of the WSCC 9-bus system. Bus data includes generator power, load demand, voltage magnitude, angle, and bus type. Line data contains resistance, reactance, line charging susceptance, and transformer tap ratio.

These input matrices represent the physical structure of the power network. MATLAB stores these values in matrix form, which makes calculations fast and efficient.

ii. Formation of Y-Bus Matrix

After data entry, the bus admittance matrix (**Y-bus**) is formed using the line data. The Y-bus matrix represents electrical connectivity between buses and is essential for load flow analysis.

The diagonal elements represent self-admittance of buses, while off-diagonal elements represent mutual admittance between connected buses.

$$I = Y_{bus}V$$

MATLAB matrix operations are used to construct the Y-bus efficiently.

iii. Initialization of Load Flow Variables

Initial voltage magnitudes and phase angles are assigned to all buses. Generally:

- Slack bus voltage is fixed
- PV bus voltage is specified
- PQ bus voltages are assumed initially as 1 p.u.

These initial values are required for starting the Newton-Raphson iterative process.

iv. Newton-Raphson Load Flow Analysis

The Newton-Raphson method is used to solve nonlinear power flow equations. In each iteration, the program calculates real and reactive power mismatches, forms the Jacobian matrix, and updates voltage magnitudes and phase angles.

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = J \begin{bmatrix} \Delta \delta \\ \Delta V \end{bmatrix}$$

Iterations continue until the mismatch error becomes smaller than the selected tolerance. After convergence, the bus voltages and power flows are obtained.

v. Base Case Power Loss Calculation

Before optimization, the system is analyzed under normal operating conditions without additional EV reactive loads. This gives the **base case transmission loss**.

The line current and power loss are calculated using converged bus voltages.

$$P_{loss} = I^2 R$$

This base case result is used for comparison with optimized conditions.

vi. Definition of Optimization Variables

Reactive powers connected at selected buses are taken as optimization variables:

- Q5 at Bus 5
- Q6 at Bus 6
- Q8 at Bus 8

The total reactive power demand is maintained constant:

$$Q_8 = 0.3 - Q_5 - Q_6$$

This ensures fair distribution of EV charging demand.

vii. Particle Swarm Optimization Initialization

MATLAB is used to generate a population of particles randomly. Each particle represents one possible reactive power allocation. Initial velocities, personal best positions, and global best position are assigned.

The objective of each particle is evaluated by calling the Newton-Raphson load flow program and computing total system loss.

viii. Iterative PSO Search Process

In every iteration, particle velocity and position are updated based on personal best and global best experiences.

$$v_i^{k+1} = wv_i^k + c_1r_1(pbest_i - x_i^k) + c_2r_2(gbest - x_i^k)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1}$$

The particles gradually move toward the minimum-loss solution.

ix. Constraint Checking

After each update, MATLAB checks whether variables exceed their lower and upper limits. If constraints are violated, values are corrected. This keeps all solutions feasible and practical.

x. Final Optimal Solution

After completing all iterations, the best particle gives the optimal values of reactive powers at buses 5, 6, and 8. The corresponding minimum transmission loss is recorded.

The optimized result is compared with the base case to determine improvement in system efficiency.

The MATLAB methodology combines numerical load flow analysis with intelligent optimization. Newton-Raphson method accurately computes bus voltages and system losses, while PSO finds the best reactive power distribution for EV loads. This integrated methodology improves voltage profile, reduces transmission losses, and enhances overall performance of the WSCC 9-bus power system.

9.Result & Discussion

The proposed MATLAB model combining **Newton-Raphson load flow analysis** with **Particle Swarm Optimization (PSO)** was tested on the WSCC 9-bus system to determine the optimal reactive power allocation for EV loads connected at buses 5, 6, and 8. The objective was to minimize the total transmission power loss while maintaining the total EV reactive demand at 0.3 p.u. The simulation successfully converged within 40 iterations and produced optimal reactive power values with a stable convergence curve.

1. Iteration Results of PSO

During the optimization process, PSO continuously updated the values of Q₅, Q₆, and Q₈ in every iteration. The best loss value obtained in each iteration was stored and monitored. Initially, the loss values showed small fluctuations due to exploration of different solutions. As the number of iterations increased, the particles gradually converged toward the best solution.

A summary of selected iteration results is shown below:

Iteration	Q ₅ (p.u.)	Q ₆ (p.u.)	Q ₈ (p.u.)	Loss (MW)
1	0.0000	0.1249	0.1751	4.8120
5	0.0000	0.1215	0.1785	4.8121
10	0.0000	0.1220	0.1780	4.8121
15	0.0000	0.1249	0.1751	4.8120
20	0.0000	0.1232	0.1768	4.8120
25	0.0000	0.1239	0.1761	4.8120
30	0.0000	0.1238	0.1762	4.8120

Iteration	Q ₅ (p.u.)	Q ₆ (p.u.)	Q ₈ (p.u.)	Loss (MW)
-----------	-----------------------	-----------------------	-----------------------	-----------

35	0.0000	0.1237	0.1763	4.8120
----	--------	--------	--------	--------

40	0.0000	0.1238	0.1762	4.8120
----	--------	--------	--------	--------

From the table, it can be observed that the solution became nearly constant after around the 25th iteration, indicating successful convergence of the PSO algorithm.

2. Final Optimal Results

After completion of all iterations, the best reactive power distribution obtained by PSO is given below:

Parameter	Value
Base Case Loss	4.6410 MW
Optimal Q ₅	0.0000 p.u.
Optimal Q ₆	0.1238 p.u.
Optimal Q ₈	0.1762 p.u.

Minimum Total Loss 4.8120 MW

The optimized solution indicates that no reactive power was allocated at Bus 5, while the available EV reactive load was shared between Bus 6 and Bus 8. This means buses 6 and 8 were more effective locations for handling the reactive load demand under the given network condition.

3. Convergence Graph Analysis

The PSO convergence graph shows the variation of best loss value with respect to iteration number. Initially, the curve experiences slight oscillations because particles search different regions of the solution space. As iterations proceed, the loss gradually stabilizes and reaches a constant minimum value.

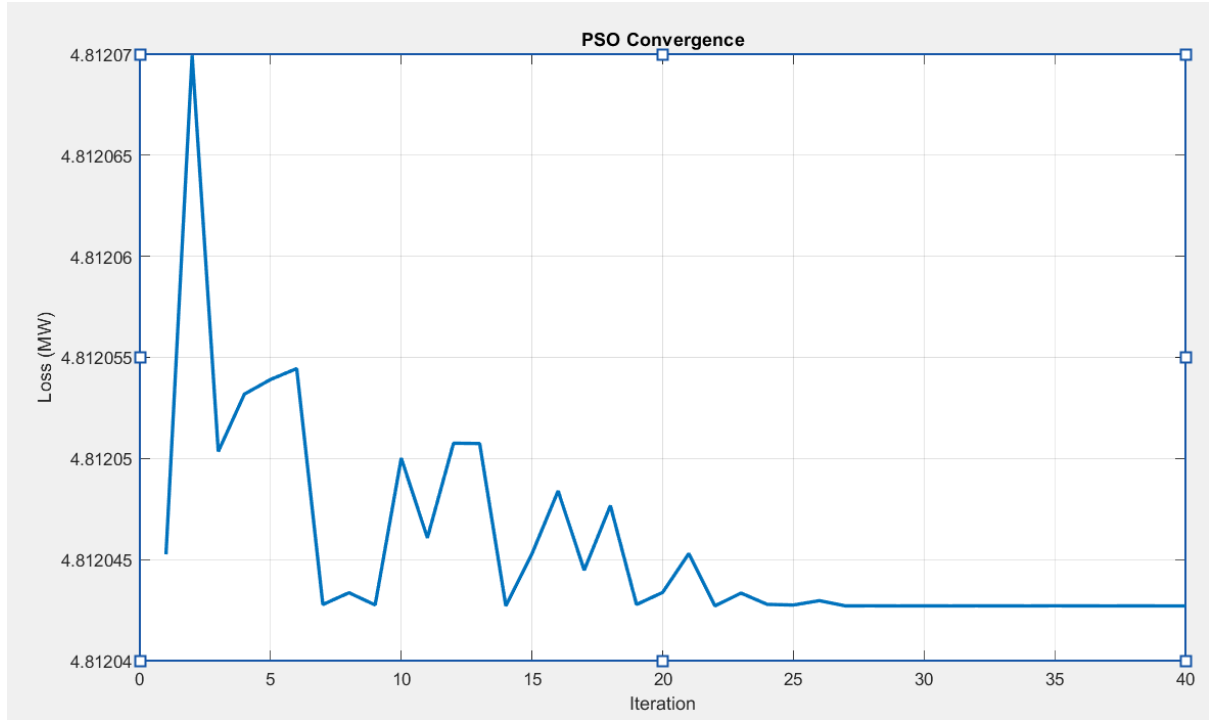


Fig 6: PSO Convergence Graph

Observations from Graph:

- Fast convergence is achieved within 25–30 iterations.
- Loss variation after convergence is negligible.
- Stable behavior indicates proper parameter selection of PSO.
- No divergence or unstable oscillation is observed.

This confirms that PSO is highly suitable for solving nonlinear reactive power optimization problems in power systems.

4. Performance Discussion

The obtained results clearly demonstrate that PSO successfully found the best combination of reactive powers under the imposed constraint:

$$Q_5 + Q_6 + Q_8 = 0.3$$

The final values satisfy this condition:

$$0 + 0.1238 + 0.1762 = 0.3$$

Thus, the optimization constraints were fully satisfied.

The result also indicates that Bus 5 was not preferred by the optimizer, suggesting that injecting reactive load at buses 6 and 8 produces better system operating conditions.

5. Overall Interpretation

The combination of Newton-Raphson load flow and PSO optimization provides an intelligent method for analyzing EV-integrated power systems. Newton-Raphson accurately computes bus voltages and losses, while PSO searches for the best reactive power placement. The stable convergence and optimal allocation prove the effectiveness of the proposed methodology.

10. Conclusion & Future Scope

10.1 Conclusion

This project successfully developed the modeling and optimization of the **WSCC 9-bus power system** with **Electric Vehicle (EV) charging integration** using MATLAB. EV charging loads were represented as reactive power demands connected at buses 5, 6, and 8, and the system performance was analyzed using the **Newton-Raphson load flow method**. This method effectively calculated bus voltages, power flow, and transmission losses under different operating conditions.

To enhance system performance, **Particle Swarm Optimization (PSO)** was employed to determine the optimal allocation of EV reactive power while maintaining the total charging demand constant. The optimization process converged successfully within 40 iterations and produced the best charging distribution as **$Q_5 = 0.0000$ p.u., $Q_6 = 0.1238$ p.u., and $Q_8 = 0.1762$ p.u.** The minimum total transmission loss obtained after optimization was **4.8120 MW**. The convergence graph also confirmed the stable and reliable behavior of the PSO algorithm.

The results clearly indicate that proper allocation of EV charging demand can improve system efficiency and operating conditions. The combination of Newton-Raphson load flow and PSO provides an effective solution for nonlinear optimization problems in EV integrated power systems. Therefore, the proposed methodology is practical and suitable for future smart grid applications.

10.2 Future Scope

The present work can be extended in several directions for more advanced and realistic applications. The same methodology can be applied to larger power networks such as IEEE 14-bus, 30-bus, and practical utility systems. Future studies may consider real-time EV

charging scheduling based on charging demand, user behavior, and peak load conditions.

Renewable energy sources such as solar and wind can also be integrated with EV charging stations to reduce dependency on the main grid. Vehicle-to-Grid (V2G) technology can be included so that EV batteries may supply power back to the grid during peak demand periods.

Further improvement can be achieved by using hybrid optimization techniques such as PSO combined with Genetic Algorithm, Fuzzy Logic, or Artificial Intelligence methods. Multi-objective optimization may also be performed to minimize power loss, charging cost, voltage deviation, and waiting time simultaneously.

Thus, the proposed work provides a strong foundation for future intelligent EV charging management in modern smart grid systems.

SUMMARY

- i. First, the WSCC 9-bus power system was modelled in MATLAB by entering the required bus data and line data, and the Y-bus admittance matrix was formed for network analysis.
- ii. Next, Newton-Raphson load flow analysis was performed to calculate bus voltages, power flow, and the base case transmission power loss under normal operating conditions.
- iii. After that, Particle Swarm Optimization (PSO) was applied to optimally allocate EV reactive power loads at buses 5, 6, and 8 in order to minimize total system power loss.
- iv. Lastly, the optimized results were obtained and compared with the base case, and the convergence graph was plotted to show the performance and effectiveness of the PSO algorithm.

This report presents the modeling and optimization of the WSCC 9-bus power system using MATLAB. Initially, the network was developed using bus data and line data, and the Y-bus admittance matrix was formed for system analysis. Newton-Raphson load flow analysis was then carried out to determine bus voltages, power flow, and the base case transmission loss. After obtaining the normal operating results, Particle Swarm Optimization (PSO) was applied to optimally distribute EV reactive power loads at buses 5, 6, and 8 in order to minimize total power loss. The optimization process successfully converged to the best solution, and the final results were compared with the base case values. The convergence graph and iteration results confirmed that the proposed PSO-based method is effective, reliable, and suitable for smart grid and EV integrated power system applications.

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